Assignment number 2 (intro to programming):

1. Write a function that takes a list of strings and a number K. The function should return a new list containing only the strings that have a length greater than K.
    - Input: list=["apple", "pie", "banana"], K=3
    - Output: ['apple', 'banana']

2. Write a function named area() that calculates the area of either a "circle" or a "rectangle". The function should take the shape name as the first argument, followed by the necessary dimensions (radius for circle; length and width for rectangle).
    - Example Input: area("circle", 5) OR area("rectangle", 4, 6)
    - Example Output: Circle Area: 78.5, Rectangle Area: 24

3. Write a program that takes an email address as input and splits it into two parts: the "username" and the "domain".
    - Example Input: support@google.com
    - Example Output: Username: support, Domain: google.com

4. Write a program to find the sum of all numbers in a list. ensure the program ignores any elements in the list that are not numbers (like strings or booleans).
    - Example Input: [10, "apple", 20, True, 5]
    - Example Output: Total Sum: 35

5. Write a Python class Restaurant with attributes like menu_items, book_table, and customer_orders, and methods like add_item_to_menu, book_tables, and customer_order. (Note: Use dictionaries and lists to store the data.)
   Perform the following tasks now:
    - Now add items to the menu.
    - Make table reservations.
    - Take customer orders.
    - Print the menu.

- Print table reservations.
- Print customer orders.

6. A Bike Rental System Let us start writing the class Customer that is composed of the following attributes:
   - name
   - age

   Here are some of the methods that you need to add to your class Customer;
   - def __init__(self).
   - Getter / setter for all attributes.

   class MainBikeRental that is composed of the following attributes:
   - List of bikes
   - List of Rentals

   Here are some of the methods that you need to add to your class MainBikeRental;
   - def __init__(self).
   - Getter / setter for all attributes.
   - requestBike: Takes a request from the customer for the number of bikes and check if the requested number is available or not if yes, function will deduct number_of_bikes from shop stock else it will print wait 10 mins. And you have to make sure that the age of the customer is over than 6 years.
   - returnBike: Allows customers to return their bikes to the rental shop by taking rentalTime, num_of_bikes as parameters. And calculate the total bill cost by multiplying rentalTime * 40. Also, add the returned bikes to the stock store again
   - addBike: adds bikes into the list
   - totalCost: returns the total cost of bikes in the store

   class Bike that is composed of the following attributes:
   - name
   - price

   Here are some of the methods that you need to add to your class Bike;
   - def __init__(self).
   - Getter / setter for all attributes.